

Interesting component: the half-bridge measurement circuit

The Li-ion Tamer gas sensor is a resistive sensor as its resistance changes in proportion to gas concentration. The challenge is converting that resistance change into a voltage signal an Arduino ADC can read cleanly.

I used a half-bridge configuration so the sensor is resistance in series with a precision reference resistor, with the midpoint voltage fed to the Arduino ADC. As gas concentration increases, sensor resistance drops, and the midpoint voltage rises proportionally. The elegance of this approach is that it sets a stable baseline when there is no gas, the divider is in a known resting state, so any deviation from that baseline is directly meaningful. It is the simplest possible way to convert an impedance change to a voltage change with minimal additional circuitry. A full Wheatstone bridge would add a second divider leg and a differential amplifier to further reject common-mode noise, but for this application the half-bridge gave sufficient signal-to-noise ratio for characterization purposes.

Feature designed for ease of use: the NinjaFlex enclosure

NinjaFlex is a flexible TPU filament. I chose it specifically because the enclosure needed to create a repeatable gas-tight seal around the sensor and inlet. A rigid enclosure would require O-rings or gaskets, and any gap would allow ambient air to dilute the test gas and corrupt the concentration measurements. The flexible material lets the enclosure compress slightly around the sensor body and inlet fitting, creating a self-sealing contact that is easy to open and reseal between experiments. This made the test protocol much faster to run because repositioning the sensor between concentration steps did not require any tools or re-sealing procedures.